

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286797

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

O; Hiroshi et al.

SERVER AND REMOTE QUALITY MEASUREMENT METHOD

Abstract

A server includes a collection unit configured to collect a response to a signal transmitted to each of a plurality of remote devices to be remotely monitored; and a comparison and determination unit configured to perform comparison and determination using a plurality of responses obtained from the plurality of remote devices at a timing when the responses have been obtained from all of the plurality of remote devices.

Inventors:	O; Hiroshi (Musashino-shi, JP), ASAKA; Kota (Musashino-shi, JP), SHIMADA; Tatsuya (Musashino-shi, JP)
Applicant:	NIPPON TELEGRAPH AND TELEPHONE CORPORATION (Tokyo, JP)
Family ID:	88518178
Assignee:	NIPPON TELEGRAPH AND TELEPHONE CORPORATION (Tokyo, JP)
Appl. No.:	18/858533
Filed (or PCT Filed):	April 25, 2022
PCT No.:	PCT/JP2022/018720

Publication Classification

Int. Cl.: H04L43/08 (20220101); H04L43/02 (20220101)

U.S. Cl.:

CPC H04L43/08 (20130101); H04L43/02 (20130101);

Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a server and a remote quality measurement method.

BACKGROUND ART

[0002] Conventionally, a technique for quality assurance control of a network in real time has been proposed. As methods of collecting quality information of a network, there are methods on a request (trigger) basis and methods on a subscription (cycle) basis. In conventional NW quality information collection, quality information is collected from a plurality of remote devices on a request (trigger) basis or a subscription (cycle) basis, and quality assurance control is performed in real time by comparing the collected quality information.

CITATION LIST

Non Patent Literature

[0003] Non Patent Literature 1: RFC1157, “A Simple Network Management Protocol (SNMP)”,
<URL: <https://datatracker.ietf.org/doc/html/rfc1157>>

[0004] Non Patent Literature 2: ITU-T, G.989.3, “40-Gigabit-capable passive optical networks (NG-PON2): Transmission convergence layer specification”

SUMMARY OF INVENTION

Technical Problem

[0005] However, the timing of receiving quality information from each remote device is not synchronized with the timing of starting comparison. Therefore, there is a problem that a deviation occurs between the two timings, and thus rapid control cannot be performed and optimum control cannot necessarily be realized.

[0006] In view of the above circumstances, an object of the present invention is to provide a technique through which optimal quality assurance control can be performed in a communication network.

Solution to Problem

[0007] An aspect of the present invention is a server including a collection unit configured to collect a response to a signal transmitted to each of a plurality of remote devices to be remotely monitored, and a comparison and determination unit configured to perform comparison and determination using a plurality of responses obtained from the plurality of remote devices at a timing when the responses have been obtained from all of the plurality of remote devices.

[0008] An aspect of the present invention is a remote quality measurement method including collecting a response to a signal transmitted to each of a plurality of remote devices to be remotely monitored, and performing comparison and determination using a plurality of responses obtained from the plurality of remote devices at a timing when responses have been obtained from all of the plurality of remote devices.

Advantageous Effects of Invention

[0009] According to the present invention, it is possible to perform optimum quality assurance control in a communication network.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. 1 A diagram illustrating a configuration example of a communication system in a first embodiment.

[0011] FIG. 2 A flowchart illustrating a flow of processing of a server in the first embodiment.

[0012] FIG. 3 A flowchart illustrating a flow of processing of a server in a second embodiment.

[0013] FIG. 4 A flowchart illustrating a flow of processing of the server in the second embodiment.

[0014] FIG. 5 A diagram illustrating a configuration example of a communication system in a third embodiment.

[0015] FIG. 6 A flowchart illustrating a flow of processing of a server in the third embodiment.

DESCRIPTION OF EMBODIMENTS

[0016] Hereinafter, embodiments of the present invention will be described with reference to drawings.

First Embodiment

[0017] FIG. 1 is a diagram illustrating a configuration example of a communication system **100** in a first embodiment. The communication system **100** includes a server **10**, a plurality of remote devices **20-1** and **20-2**, a plurality of switches **30-1** to **30-3**, and a terminal **40**. The number of remote devices **20** and the number of switches **30** are not limited to the numbers illustrated in FIG. 1. Hereinafter, the remote devices **20-1** and **20-2** will be referred to as the remote device **20** in a case in which they are not distinguished, and the switches **30-1** to **30-3** will be referred to as the switch **30** in a case in which they are not distinguished.

[0018] In the example illustrated in FIG. 1, the server **10** is connected to the switches **30-2** and **30-3** via a transmission line. The switch **30-2** is connected to the switch **30-1** and the remote device **20-1** via a transmission line. The switch **30-3** is connected to the switch **30-1** and the remote device **20-2** via a transmission line. The switch **30-1** is connected to the switches **30-2** and **30-3** and the terminal **40** via a transmission line. A transmission line may be an optical transmission line such as an optical fiber or an electric line such as a coaxial cable. In the following description, it is assumed that a transmission line is an optical transmission line. Note that the remote devices **20-1** and **20-2** may be connected to the server **10** via the switch **30** through the same line.

[0019] The server **10** collects quality information from each of the remote devices **20-1** and **20-2**. For example, the server **10** transmits a request or a subscribe signal to the remote devices **20-1** and **20-2**. The server **10** receives reports transmitted from the remote devices **20-1** and **20-2**. A report is a response to the request or subscribe signal. The server **10** performs comparison and determination using the collected quality information and controls a path and the remote devices **20-1** and **20-2** on the basis of the determination result. The server **10** is an aspect of a monitoring device.

[0020] The remote device **20** is a device to be monitored in a network. Upon receiving the request or subscribe signal transmitted from the server **10**, the remote device **20** transmits a report to the server **10**.

[0021] The switch **30** switches a route according to control of the server **10**. The switch **30** connects the server **10** and the remote device **20** such that they can communicate or connects the server **10** and the terminal **40** such that they can communicate according to route switching. Although the switch **30** will be described as an optical switch below, the switch **30** may be an electrical switch such as an L2SW (L2 switch) or an L3SW (L3 switch), or a router in a case where electrical communication is performed.

[0022] The terminal **40** is a communication device for communicating with the server **10**.

[0023] Next, a specific configuration of the server **10** will be described. The server **10** includes a communication unit **11**, a collection unit **12**, a comparison and determination unit **13**, and a control unit **14**. Note that it is assumed in the first embodiment that the comparison and determination unit **13** does not perform processing until there is an instruction from another functional unit (for example, the collection unit **12**). That is, in the first embodiment, the comparison and determination unit **13** is a functional unit that does not operate autonomously. Processing performed by the comparison and determination unit **13** is, for example, quality comparison and determination.

[0024] The communication unit **11** communicates with the remote device **20** via the switch **30**. The communication unit **11** transmits a request or a subscribe signal to the remote device **20** and receives a report from the remote device **20**.

[0025] The collection unit **12** collects reports transmitted from respective remote devices **20**. The

collection unit **12** in the first embodiment causes the comparison and determination unit **13** to execute processing in a case in which reports have been received from all the remote devices **20**. Specifically, in a case in which the collection unit **12** has received reports from all the remote devices **20**, the collection unit **12** outputs a trigger (hereinafter referred to as an “activation signal”) for causing the comparison and determination unit **13** to execute processing to the comparison and determination unit **13**, thereby causing the comparison and determination unit **13** to execute processing. Here, it is assumed that all the remote devices **20** are remote devices **20** whose quality information is to be collected in the server **10** and are registered in advance. The collection unit **12** provides a reception flag for each registered remote device **20** and determines that reports have been received from all the remote devices **20** in a case in which all reception flags are TRUE.

[0026] The comparison and determination unit **13** performs processing according to the activation signal output from the collection unit **12**. Specifically, when the activation signal is obtained from the collection unit **12**, the comparison and determination unit **13** acquires the reports transmitted from the respective remote devices **20** from collection unit **12** and performs quality comparison and determination using the acquired reports. Note that processing performed by the comparison and determination unit **13** is the same as conventional processing. The comparison and determination unit **13** performs comparison and determination, and in a case in which it is determined that control is necessary, notifies the control unit **14** of details of control. On the other hand, the comparison and determination unit **13** performs comparison and determination, and in a case in which it is determined that control is not necessary, does nothing.

[0027] The control unit **14** generates a request or subscribe signal. The control unit **14** controls the communication unit **11** to transmit the generated request or subscribe signal to the remote device **20**. The control unit **14** controls the remote device **20** and the switch **30** in response to details of notification from the comparison and determination unit **13**.

[0028] FIG. 2 is a flowchart illustrating a flow of processing of the server **10** in the first embodiment. Note that it is assumed that the server **10** and the remote device **20** are connected via the switch **30** such that they can communicate at the start of processing in FIG. 2.

[0029] The collection unit **12** initializes the reception flags corresponding to all the remote devices **20** to FALSE (step S101). The control unit **14** generates a request addressed to each remote device **20**. The control unit **14** transmits the request to each remote device **20** via the communication unit **11** (step S102). Note that, although the configuration in which the server **10** transmits a request to each remote device **20** has been described here, the server **10** may transmit a subscribe signal to each remote device **20**.

[0030] The request transmitted from the server **10** is received by the remote device **20-1** via the switch **30-2**. The request transmitted from the server **10** is received by the remote device **20-2** via the switch **30-3**. The collection unit **12** determines whether a report has been received (step S103). If it is determined that a report has not been received (NO in step S103), the collection unit **12** waits until a report is received.

[0031] On the other hand, if it is determined that a report has been received (YES in step S103), the collection unit **12** changes the reception flag of the remote device **20** that is the transmission source of the report (for example, the remote device **20-1**) from FALSE to TRUE (step S104). Thereafter, the collection unit **12** determines whether or not all the reception flags are TRUE (step S105). If it is determined that not all the reception flags are TRUE (some or all of the reception flags are FALSE) (NO in step S105), the collection unit **12** waits until a further report is received.

[0032] As an example, in the reception flags of the collection unit **12**, only the reception flag corresponding to the remote device **20-1** is TRUE, and only the reception flag corresponding to the remote device **20-2** is FALSE in the above-described example. Therefore, the collection unit **12** determines that not all the reception flags are TRUE.

[0033] Thereafter, when a report transmitted from the remote device **20-2** is received in processing of step S103, the collection unit **12** changes the reception flag of the remote device **20** that is the

transmission source of the report (for example, the remote device **20-2**) from FALSE to TRUE. The collection unit **12** determines whether or not all the reception flags are TRUE. In this case, in the reception flags of the collection unit **12**, the reception flag corresponding to the remote device **20-1** and the reception flag corresponding to the remote device **20-2** are TRUE. Therefore, the collection unit **12** determines that all the reception flags are TRUE.

[0034] As described above, if it is determined that all the reception flags are TRUE (YES in step **S105**), the collection unit **12** outputs an activation signal to the comparison and determination unit **13**. The collection unit **12** may reset all the reception flags to FALSE after outputting the activation signal.

[0035] When the activation signal is obtained from the collection unit **12**, the comparison and determination unit **13** acquires the report transmitted from each remote device **20** from the collection unit **12**. The comparison determination unit **13** performs comparison determination using the acquired report of each remote device **20** (step **S106**). As a result of comparison and determination, the comparison and determination unit **13** determines whether or not control is necessary (step **S107**). If it is determined that control is not necessary (NO in step **S107**), the comparison and determination unit **13** ends processing.

[0036] On the other hand, if it is determined that control is necessary (YES in step **S107**), the comparison and determination unit **13** notifies the control unit **14** of details of control (step **S108**). The control unit **14** generates a control signal for controlling the remote device **20** or the switch **30** on the basis of the details of control notified of by the comparison and determination unit **13**. The control unit **14** performs control by transmitting the generated control signal to the remote device **20** or the switch **30** via the communication unit **11** (step **S109**).

[0037] According to the communication system **100** configured as described above, the comparison and determination unit **13** is caused to execute comparison and determination processing at the timing when the collection unit **12** has received reports from all remote devices **20**. Accordingly, the timing of receiving quality information from each remote device **20** and the timing of starting comparison and determination are aligned. Therefore, optimum quality assurance control in a communication network can be performed.

Second Embodiment

[0038] In the first embodiment, the configuration in which the comparison and determination unit operates according to the activation signal from the collection unit has been described. In a second embodiment, a configuration in which the comparison and determination unit autonomously operates will be described.

[0039] A system configuration in the second embodiment is similar to that in the first embodiment. Furthermore, functional units included in the server **10** are similar to those in the first embodiment except that the comparison and determination unit **13** autonomously operates and the activation signal is not output from the collection unit **12**. Here, in a case in which the comparison and determination unit **13** autonomously operates, the comparison and determination unit **13** reads the reception flags held by the collection unit **12** at a high speed without synchronizing with processing of the collection unit **12**.

[0040] FIG. **3** is a flowchart illustrating a flow of processing of the server **10** in the second embodiment. Note that it is assumed that the server **10** and the remote device **20** are connected via the switch **30** such that they can communicate at the start of processing of FIG. **3**. In FIG. **3**, processing performed by the collection unit **12** will be mainly described.

[0041] The collection unit **12** initializes the reception flags corresponding to all the remote devices **20** to FALSE (step **S201**). The control unit **14** generates a request addressed to each remote device **20**. The control unit **14** transmits the request to each remote device **20** via the communication unit **11** (step **S202**). Note that, although the configuration in which the server **10** transmits a request to each remote device **20** has been described here, the server **10** may transmit a subscribe signal to each remote device **20**.

[0042] The request transmitted from the server **10** is received by the remote device **20-1** via the switch **30-2**. The request transmitted from the server **10** is received by the remote device **20-2** via the switch **30-3**. The collection unit **12** determines whether a report has been received (step **S203**). If it is determined that a report has not been received (NO in step **S203**), the collection unit **12** waits until a report is received.

[0043] On the other hand, if it is determined that a report has been received (YES in step **S203**), the collection unit **12** changes the reception flag of the remote device **20** (for example, remote device **20-1**) that is the transmission source of the report from FALSE to TRUE (step **S204**). Thereafter, the collection unit **12** repeatedly executes processing of step **S203** and subsequent processing until reports are obtained from all the remote devices **20**.

[0044] FIG. **4** is a flowchart illustrating a flow of processing of the server **10** in the second embodiment. In FIG. **4**, processing performed by the comparison and determination unit **13** will be mainly described.

[0045] The comparison and determination unit **13** reads reception flags of the collection unit **12** at a high speed (step **S211**). The comparison and determination unit **13** determines whether or not all reception flags are TRUE with reference to the read reception flags of the collection unit **12** (step **S212**). If the comparison and determination unit **13** determines that not all the reception flags are TRUE (NO in step **S212**), the comparison and determination unit **13** reads the reception flags of the collection unit **12** again at a high speed.

[0046] On the other hand, if it is determined that all the reception flags are TRUE (YES in step **S212**), the comparison and determination unit **13** resets all the reception flags of the collection unit **12** to FALSE (step **S213**). The comparison and determination unit **13** acquires the report of each remote device **20** from the collection unit **12** and performs comparison determination using the acquired report of each remote device **20** (step **S214**). As a result of comparison and determination, the comparison and determination unit **13** determines whether or not control is necessary (step **S215**). If it is determined that control is not necessary (NO in step **S215**), the comparison and determination unit **13** ends processing.

[0047] On the other hand, if it is determined that control is necessary (YES in step **S215**), the comparison and determination unit **13** notifies the control unit **14** of details of control (step **S216**). The control unit **14** generates a control signal for controlling the remote device **20** or the switch **30** on the basis of the details of control notified of by the comparison and determination unit **13**. The control unit **14** performs control by transmitting the generated control signal to the remote device **20** or the switch **30** via the communication unit **11** (step **S217**).

[0048] According to the communication system **100** of the second embodiment configured as described above, the comparison and determination unit **13** autonomously monitors reception flags of the collection unit **12** at a high speed, and in a case in which the reception flags corresponding to all the remote devices **20** become TRUE (a state indicating that the reports have been received from all the remote devices **20**), performs comparison and determination using the reports obtained from all the remote devices **20** from the collection unit **12**. Accordingly, the timing of receiving quality information from each remote device **20** and the timing of starting comparison and determination are aligned. Therefore, optimum quality assurance control in a communication network can be performed.

Third Embodiment

[0049] In a third embodiment, a configuration in which the comparison and determination unit is caused to execute processing in accordance with a time at which a report is predicted to be received will be described.

[0050] FIG. **5** is a diagram illustrating a configuration example of a communication system **100b** in the third embodiment. The communication system **100b** includes a server **10b**, a plurality of remote devices **20-1** and **20-2**, a plurality of switches **30-1** to **30-3**, and a terminal **40**.

[0051] The communication system **100b** is different from the communication system **100** in that

the server **10b** is provided instead of the server **10**. Other components of the communication system **100b** are similar to those of the communication system **100**. Hereinafter, differences from the communication system **100** will be described.

[0052] The server **10b** includes a communication unit **11**, a collection unit **12b**, a comparison and determination unit **13**, a control unit **14**, and a timing adjustment unit **15**. Note that, similarly to the first embodiment, it is assumed that the comparison and determination unit **13** does not perform processing until there is an instruction from another functional unit (for example, the timing adjustment unit **15**). That is, in the third embodiment, the comparison and determination unit **13** is a functional unit that does not autonomously operate.

[0053] The collection unit **12b** collects a report transmitted from each remote device **20**. In a case in which the collection unit **12b** has received the reports from all the remote devices **20**, the collection unit **12b** acquires a time (hereinafter referred to as “reception completion time”) at which all the reception flags become TRUE. The collection unit **12b** notifies the timing adjustment unit **15** of the acquired information on the reception completion time.

[0054] The timing adjustment unit **15** estimates a predicted next report reception time on the basis of information on the past reception completion time notified of by the collection unit **12b**. In a case in which a report is periodically transmitted, for example, the timing adjustment unit **15** calculates a difference between two past reception completion times and calculates a time obtained by adding a difference from the last report reception time to the difference as a predicted report reception time. The timing adjustment unit **15** causes the comparison and determination unit **13** to execute processing in accordance with the predicted report reception time. Note that, in a case in which it takes time to process the report received by the collection unit **12b**, the timing adjustment unit **15** may consider the time to process the report at the predicted report reception time (for example, add the time to process the report to the predicted report reception time).

[0055] FIG. **6** is a flowchart illustrating a flow of processing of the server **10b** in the third embodiment. Note that it is assumed that the server **10b** and the remote device **20** are connected via the switch **30** such that they can communicate at the start of processing of FIG. **6**. In FIG. **6**, processing similar to that in FIG. **2** is denoted by the same reference numeral as that in FIG. **2**, and description thereof is omitted.

[0056] Processing from step **S101** to step **S105** is executed, and if it is determined that all the reception flags are TRUE (YES in step **S105**), the collection unit **12b** acquires a reception completion time. The collection unit **12b** notifies the timing adjustment unit **15** of the acquired reception completion time.

[0057] The timing adjustment unit **15** estimates a predicted next report reception time on the basis of the past reception completion time notified of by the collection unit **12b** (step **S301**). Thereafter, the timing adjustment unit **15** determines whether or not the current time is the predicted report reception time (step **S302**). If it is determined that the current time is not the predicted report reception time (NO in step **S302**), the timing adjustment unit **15** waits until the current time reaches the predicted report reception time.

[0058] On the other hand, if it is determined that the current time is the predicted report reception time (YES in step **S302**), the timing adjustment unit **15** causes the comparison and determination unit **13** to execute processing (step **S303**). Specifically, the timing adjustment unit **15** causes the comparison and determination unit **13** to execute the processing by outputting an activation signal to the comparison and determination unit **13** at the timing when the current time reaches the predicted report reception time. Thereafter, processing of step **S106** and subsequent steps are executed.

[0059] According to the communication system **100b** configured as described above, a predicted next report reception time is estimated on the basis of the time at which reception of reports from all the remote devices **20** is completed (reception completion time). Then, the timing adjustment unit **15** causes the comparison and determination unit **13** to execute comparison and determination

processing in accordance with the predicted report reception time. The comparison and determination unit **13** performs comparison and determination at a timing when a report is received. Accordingly, the timing of receiving quality information from each remote device **20** and the timing of starting comparison and determination are aligned. Therefore, optimum quality assurance control in a communication network can be performed.

[0060] Some functional units of the servers **10** and **10b** and the remote device **20** in the above-described embodiments may be realized by a computer. In that case, a program for realizing these functions may be recorded in a computer-readable recording medium, and the functions may be realized by causing a computer to read and execute the program recorded in this recording medium. Note that the “computer system” mentioned herein includes an OS and hardware such as peripheral devices.

[0061] In addition, the “computer-readable recording medium” refers to a portable medium such as a flexible disk, a magneto-optical disc, a ROM, or a CD-ROM or a storage device such as a hard disk included in the computer system. Further, the “computer-readable recording medium” may include a medium that dynamically holds the program for a short time, such as a communication line in a case where the program is transmitted via a network such as the Internet or a communication channel such as a telephone line, and a medium that holds the program for a certain period of time, such as a volatile memory inside a computer system serving as a server or a client in that case. In addition, the aforementioned program may be for realizing some of the functions described above, may realize the functions described above by a combination with a program already recorded in the computer system, or may be realized using a programmable logic device such as an FPGA.

[0062] Although the embodiments of the present invention has been described in detail with reference to the drawings, the specific configuration is not limited to these embodiments and includes design and the like within a range not departing from the gist of the present invention.

INDUSTRIAL APPLICABILITY

[0063] The present invention can be applied to a communication system that collects quality information of a network.

REFERENCE SIGNS LIST

[0064] **10, 10b** Server [0065] **20, 20-1 to 20-2** Remote device [0066] **30, 30-1 to 30-3** Switch [0067] **40** Terminal [0068] **11** Communication unit [0069] **12, 12b** Collection unit [0070] **13** Comparison and determination unit [0071] **14** Control unit [0072] **15** Timing adjustment unit

Claims

1. A server comprising: a collector configured to collect a response to a signal transmitted to each of a plurality of remote devices to be remotely monitored; and a comparison and determiner configured to perform comparison and determination using a plurality of responses obtained from the plurality of remote devices at a timing when the responses have been obtained from all of the plurality of remote devices.
2. The server according to claim 1, wherein the collector holds a flag indicating whether or not a response has been obtained for each remote device, each time a response is obtained from each remote device, changes a state of the flag corresponding to a remote device from which a response has been obtained to a state indicating that the response has been obtained, and outputs an activation signal for causing the comparison and determiner to execute processing to the comparison and determiner at a timing when the state of the flag has become a state indicating that responses have been obtained from all of the plurality of remote devices, and the comparison and determiner performs comparison and determination using a plurality of responses obtained from all of the plurality of remote devices when the activation signal is acquired.
3. The server according to claim 1, wherein the collector holds a flag indicating whether or not a

response has been obtained for each remote device, each time a response is obtained from each remote device, changes a state of the flag corresponding to a remote device from which a response has been obtained to a state indicating that the response has been obtained, and the comparison and determiner monitors a flag in the collector and performs comparison and determination using a plurality of responses obtained from all of the plurality of remote devices at a timing when a state of the flag has become a state indicating that responses have been obtained from all of the plurality of remote devices.

4. The server according to claim 1, further comprising a timing adjuster configured to predict a time at which responses are received from the remote devices, wherein the collector holds a flag indicating whether or not a response has been obtained for each remote device, each time a response is obtained from each remote device, changes a state of the flag corresponding to a remote device from which a response has been obtained to a state indicating that the response has been obtained, and notifies the timing adjuster of a time when the state of the flag has become a state indicating that responses have been obtained from all of the plurality of remote devices as a reception completion time, the timing adjuster calculates a predicted reception time that is a time at which a response is received next from the remote device on the basis of information on the reception completion time, and outputs an activation signal for causing the comparison and determiner to execute processing to the comparison and determiner at the calculated predicted reception time, and the comparison and determiner performs comparison and determination using a plurality of responses obtained from all of the plurality of remote devices when the activation signal is acquired.

5. The server according to claim 4, wherein, in a case in which a time is required for processing of a response in the collector, the timing adjuster adds the time required for processing of a response to the predicted reception time and outputs the activation signal to the comparison and determiner.

6. A remote quality measurement method comprising: collecting a response to a signal transmitted to each of a plurality of remote devices to be remotely monitored; and performing comparison and determination using a plurality of responses obtained from the plurality of remote devices at a timing when responses have been obtained from all of the plurality of remote devices.
